Started on	Tuesday, 24 March 2026, 9:31 AM
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Time taken	4 secs
Grade	**0.00** out of 100.00

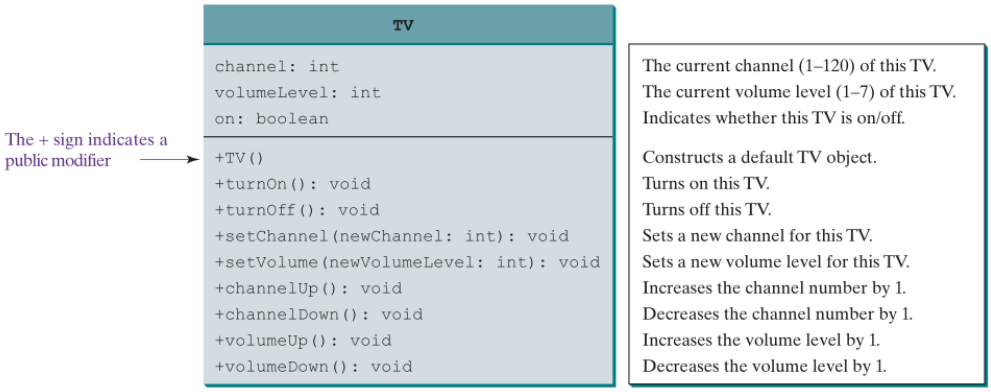
Question 1

Not answered

Mark 0.00 out of 25.00

Consider television sets. Each TV is an object with states (current channel, current volume level, and power on or off) and behaviors (change channels, adjust volume, and turn on/off). You can use a class to model TV sets.

The **UML** diagram for the class is shown as follows:



Answer: (penalty regime: 0 %)

```
1
```

▼ **Show/hide question author's solution (Java)**

```java
public class TV {
 int channel = 1; // Default channel is 1
 int volumeLevel = 1; // Default volume level is 1
 boolean on = false; // TV is off

 public TV() {
 }

 public void turnOn() {
    on = true;
 }

 public void turnOff() {
    on = false;
 }

 public void setChannel(int newChannel) {
 if (on && newChannel >= 1 && newChannel <= 120)
    channel = newChannel;
 }

 public void setVolume(int newVolumeLevel) {
 if (on && newVolumeLevel >= 1 && newVolumeLevel <= 7)
    volumeLevel = newVolumeLevel;
 }

 public void channelUp() {
 if (on && channel < 120)
    channel++;
```

```
30  }
31
32  public void channelDown() {
33  if (on && channel > 1)
34     channel--;
35  }
36
37  public void volumeUp() {
38  if (on && volumeLevel < 7)
39     volumeLevel++;
40  }
41
42  public void volumeDown() {
43  if (on && volumeLevel > 1)
44     volumeLevel--;
45  }
46 }
```

Question 2

Not answered

Mark 0.00 out of 25.00

Design a class named Rectangle to represent a rectangle. The class contains:

■ Two double data fields named width and height that specify the width and height of the rectangle. The default values are 1 for both width and height.

■ A no-arg constructor that creates a default rectangle.

■ A constructor that creates a rectangle with the specified width and height.

■ A method named getArea() that returns the area of this rectangle.

■ A method named getPerimeter() that returns the perimeter.

Optional: Write a test program that creates two Rectangle objects-one with width 4 and height 40, and the other with width 3.5 and height 35.9. Display the width, height, area, and perimeter of each rectangle in this order.

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
/*public class Exercise09_01 {
  public static void main(String[] args) {
    MyRectangle myRectangle = new MyRectangle(4, 40);
    System.out.println("The area of a rectangle with width " +
      myRectangle.width + " and height " +
      myRectangle.height + " is " +
      myRectangle.getArea());
    System.out.println("The perimeter of a rectangle is " +
      myRectangle.getPerimeter());

    MyRectangle yourRectangle = new MyRectangle(3.5, 35.9);
    System.out.println("The area of a rectangle with width " +
      yourRectangle.width + " and height " +
      yourRectangle.height + " is " +
      yourRectangle.getArea());
    System.out.println("The perimeter of a rectangle is " +
      yourRectangle.getPerimeter());
  }
}*/

class MyRectangle {
  // Data members
  double width = 1;
  double height = 1;

  // Constructor
  public MyRectangle() {
  }

  // Constructor
  public MyRectangle(double newWidth, double newHeight) {
    width = newWidth;
    height = newHeight;
  }

  public double getArea() {
    return width * height;
  }
```

```
39
40    public double getPerimeter() {
41      return 2 * (width + height);
42    }
43  }
```

Question 3

Not answered

Mark 0.00 out of 25.00

Design a class named **Stock** that contains:

■ A string data field named symbol for the stock's symbol.

■ A string data field named name for the stock's name.

■ A double data field named previousClosingPrice that stores the stock price for the previous day.

■ A double data field named currentPrice that stores the stock price for the current time.

■ A constructor that creates a stock with the specified symbol and name.

■ A method named getChangePercent() that returns the percentage changed from previousClosingPrice to currentPrice.

Optional: Write a test program that creates a Stock object with the stock symbol ORCL, the name Oracle Corporation, and the previous closing price of 34.5. Set a new current price to 34.35 and display the price-change percentage.

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
/*public class Exercise09_02 {
    public static void main(String[] args) {
      Stock stock = new Stock("SUNW", "Sun MicroSystems Inc.");
      stock.setPreviousClosingPrice(100);

      // Set current price
      stock.setCurrentPrice(90);

      // Display stock info
      System.out.println("Previous Closing Price: " +
        stock.getPreviousClosingPrice());
      System.out.println("Current Price: " +
        stock.getCurrentPrice());
      System.out.println("Price Change: " +
        stock.getChangePercent() * 100 + "%");
    }
}*/

class Stock {
  String symbol;
  String name;
  double previousClosingPrice;
  double currentPrice;

  public Stock() {
  }

  public Stock(String newSymbol, String newName) {
    symbol = newSymbol;
    name = newName;
  }

  public double getChangePercent() {
    return (currentPrice - previousClosingPrice) /
      previousClosingPrice;
  }
```

```
37
38  public double getPreviousClosingPrice() {
39    return previousClosingPrice;
40  }
41
42  public double getCurrentPrice() {
43    return currentPrice;
44  }
45
46  public void setCurrentPrice(double newCurrentPrice) {
47    currentPrice = newCurrentPrice;
48  }
49
50  public void setPreviousClosingPrice(double newPreviousClosingPrice) {
51    previousClosingPrice = newPreviousClosingPrice;
52  }
```

Question 4

Not answered

Mark 0.00 out of 25.00

Design a class named Fan to represent a fan. The class contains:

■ Three constants named SLOW, MEDIUM, and FAST with the values 1, 2, and 3 to denote the fan speed.

■ A private int data field named speed that specifies the speed of the fan (the default is SLOW).

■ A private boolean data field named on that specifies whether the fan is on (the default is false).

■ A private double data field named radius that specifies the radius of the fan (the default is 5).

■ A string data field named color that specifies the color of the fan (the default is blue).

■ The accessor and mutator methods for all four data fields.

■ A no-arg constructor that creates a default fan.

■ A method named toString() that returns a string description for the fan. If the fan is on, the method returns the fan speed, color, and radius in one combined string. If the fan is not on, the method returns the fan color and radius along with the string "fan is off" in one combined string.

Optional: Write a test program that creates two Fan objects. Assign maximum speed, radius 10, color yellow, and turn it on to the first object. Assign medium speed, radius 5, color blue, and turn it off to the second object. Display the objects by invoking their **toString** method.

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
/*public class Exercise09_08 {
  public static void main(String[] args) {
    Fan1 fan1 = new Fan1();
    fan1.setSpeed(Fan1.FAST);
    fan1.setRadius(10);
    fan1.setColor("yellow");
    fan1.setOn(true);
    System.out.println(fan1.toString());

    Fan1 fan2 = new Fan1();
    fan2.setSpeed(Fan1.MEDIUM);
    fan2.setRadius(5);
    fan2.setColor("blue");
    fan2.setOn(false);
    System.out.println(fan2.toString());
  }
}*/

class Fan1 {
  public static int SLOW = 1;
  public static int MEDIUM = 2;
  public static int FAST = 3;

  private int speed = SLOW;
  private boolean on = false;
  private double radius = 5;
  private String color = "white";

  public Fan1() {
  }

  public int getSpeed() {
```

```
33    return speed;
34  }
35
36  public void setSpeed(int newSpeed) {
37    speed = newSpeed;
38  }
39
40  public boolean isOn() {
41    return on;
42  }
43
44  public void setOn(boolean trueOrFalse) {
45    this.on = trueOrFalse;
46  }
47
48  public double getRadius() {
49    return radius;
50  }
51
52  public void setRadius(double newRadius) {
```